

MOHAMMED ADILSHA AFSAR M

Electronics & Communication Engineer | Embedded Systems Enthusiast | IoT & Robotics Developer

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OBJECTIVE

Electronics and Communication Engineering graduate with practical experience in embedded systems, robotics, IoT, networking, and technical support. Skilled in developing hardware based solutions using Raspberry Pi, Arduino, ESP32, and microcontrollers. Experienced in robotics training, embedded programming, troubleshooting, industrial documentation, and electronics project development. Passionate about Embedded Systems, Embedded Linux, IoT, Automation, and Robotics, with a strong interest in building reliable real world electronic systems.

EDUCATION

College of Engineering Vadakara, Kerala, APJ Kerala Technological University (2020-2024)
Bachelor of Technology(B.Tech) in **Electronics and Communication Engineering**

TECHNICAL SKILLS

• Programming

- C
- Python
- Embedded C (Learning)
- Arduino Programming

• Embedded Systems

- Raspberry Pi
- Arduino UNO
- ESP8266 / NodeMCU
- ESP32
- GPIO Programming
- UART
- I2C
- SPI
- PWM

• Electronics

- Analog Electronics
- Digital Electronics
- Electronic Circuit Design
- Sensors & Actuators
- Motor Drivers
- Servo Control
- Power Supply Basics
- PCB Assembly & Testing

• IoT & Robotics

- IoT System Development
- Raspberry Pi Projects
- Arduino Automation
- Robotics Integration
- Wireless Communication
- Sensor Interfacing

• Software & Tools

- Arduino IDE
- VS Code
- Git & GitHub
- Linux Basics
- Raspberry Pi OS
- OpenCV
- YOLO (Basic)
- MATLAB (Academic)

• Networking

- LAN Configuration
- Router Configuration
- Network Troubleshooting
- IP Addressing
- CCTV Systems

EXPERIENCE

Technical Staff & Robotics Trainer, BenchMark International School, Tirur (July 2024 - 2026)

Tirur

- Delivered Robotics and STEM training sessions to school students.
- Managed computer laboratories and IT infrastructure.
- Installed and maintained networking equipment.
- Configured LAN and internet connectivity.
- Installed and maintained CCTV systems.
- Provided technical support for classrooms and digital learning.
- Maintained electronic devices and laboratory equipment.
- Assisted in automation-related educational activities.
- Performed hardware troubleshooting and preventive maintenance.
- Supported hybrid learning technology implementation

INTERNSHIP

KELTRON (Kerala State Electronics Development Corporation Ltd.)

Intern – Electronics & IoT

May 2023

Kochi

- Learned IoT system development.
- Worked with Arduino-based automation.
- Developed embedded hardware prototypes.
- Performed sensor interfacing.
- Studied industrial electronics applications.

TECHNICAL PROJECTS

SMART AUTONOMOUS SHOPPING TROLLEY WITH ROBOTIC ARM

Major Project

- **Technologies**
Raspberry Pi 4, Arduino, Python, OpenCV, Ultrasonic Sensors, Servo Motors, DC Motors & Pi Camera
- **Responsibilities**
 - Designed an autonomous shopping trolley.
 - Integrated robotic arm for object pickup.
 - Developed object detection using computer vision.
 - Implemented obstacle avoidance.
 - Programmed motor control.
 - Integrated sensor-based navigation.
 - Developed hardware-software communication.

IOT AUTOMATION PROJECTS

- Home automation prototypes
- Sensor monitoring systems
- Microcontroller-based automation
- Wireless IoT communication

ROBOTICS PROJECTS

- Autonomous robot development
- Servo motor control systems
- Raspberry Pi robotics
- Sensor based navigation

ORGANIZATIONAL EXPERIENCE

Chief Creative Officer, Innovation and Entrepreneurship Development Centre (IEDC)

College of Engineering Vadakara

Robotics Head, Cybroque Technologies

College of Engineering Vadakara

CAREER OBJECTIVE

To build a successful career in Embedded Systems, Firmware Development, Robotics, and IoT by applying strong electronics fundamentals and continuously developing expertise in Embedded C, Embedded Linux, RTOS, and modern hardware technologies while contributing to innovative engineering solutions.
